

CleanCore AI Engine

Industrial Bin Fill-Level Classifier · Model Performance Report

0.971

mAP50

80.8%

Precision

92.9%

Recall

42

Best Epoch

ARCHITECTURE

YOLOv8s

CLASSES

4 fill-level categories

PARAMETERS

11.1M

TRAINING

735 train / 120 val / 40 test

DATASET

895 proprietary IoT images

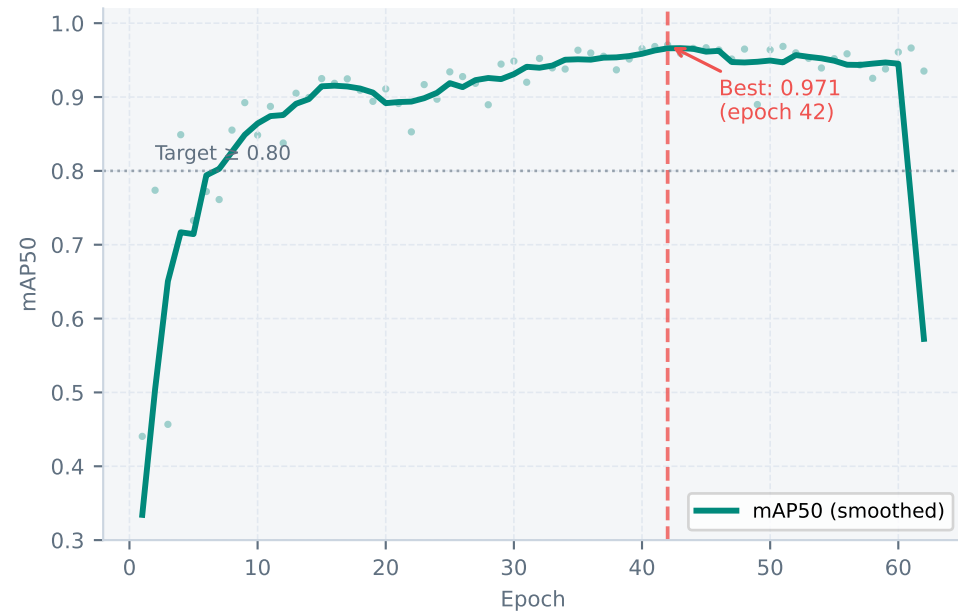
BACKBONE

COCO pretrained (transfer learning)

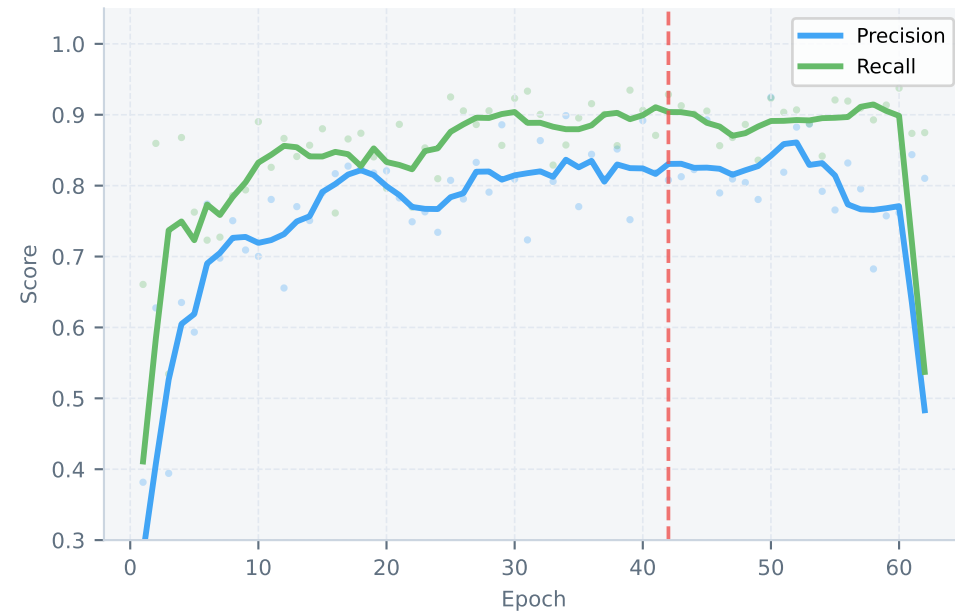
Domain: Large-scale industrial steel skip bins · Fixed IoT cameras · Real-world proprietary dataset

Training Dynamics

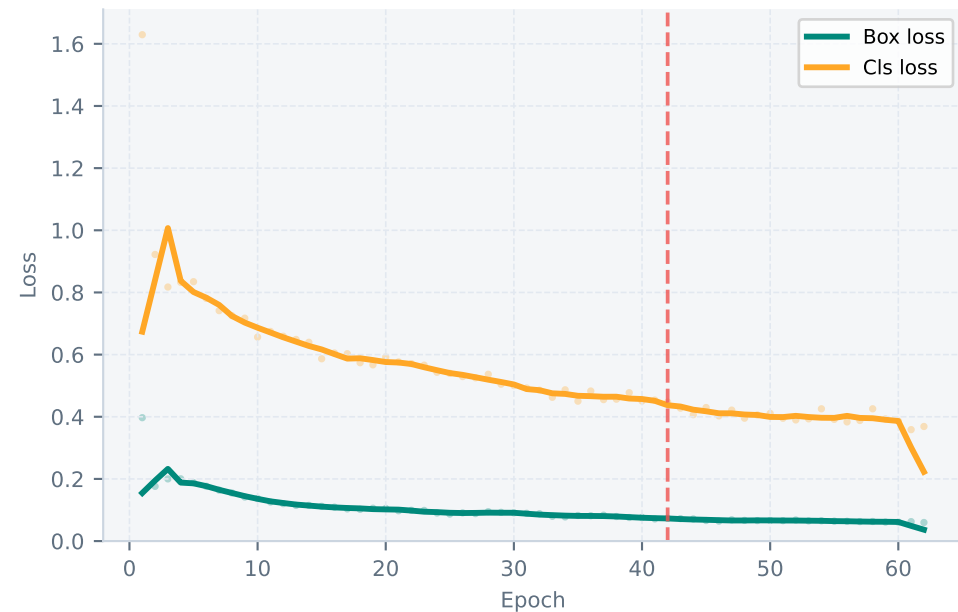
mAP50 Over Training



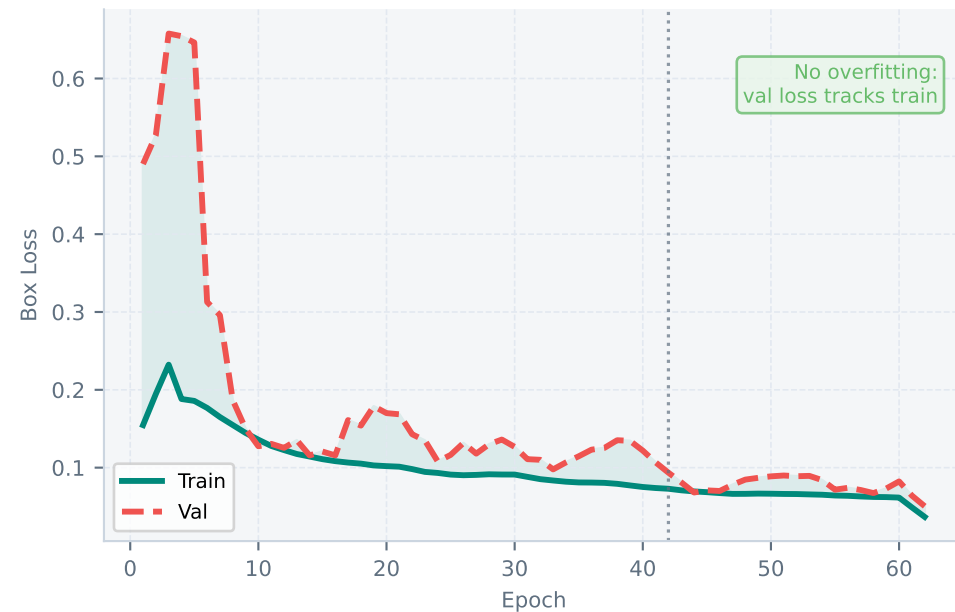
Precision & Recall Over Training



Training Loss

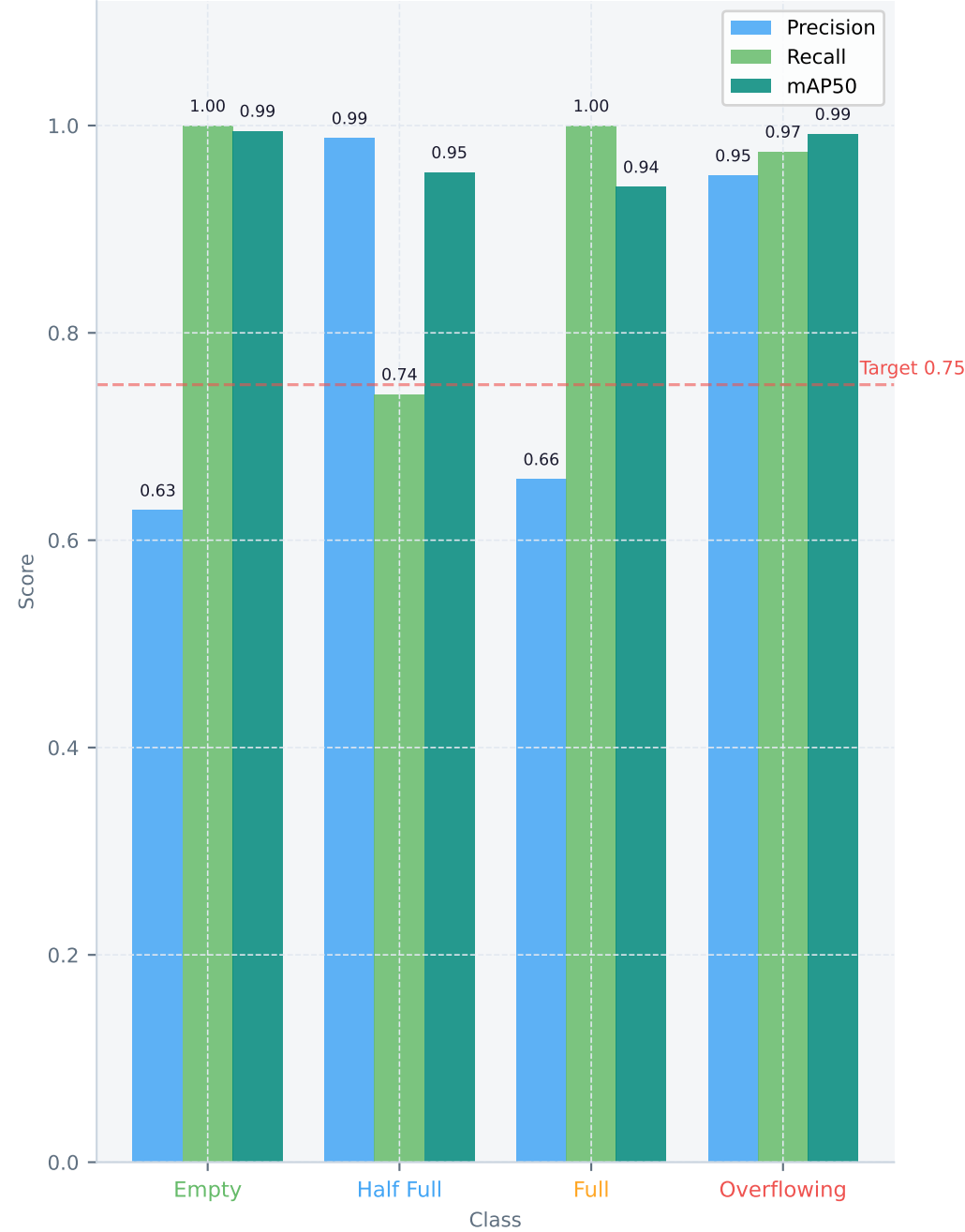


Train vs Val Box Loss (Generalisation)



Per-Class Performance Analysis

Precision / Recall / mAP50 by Class

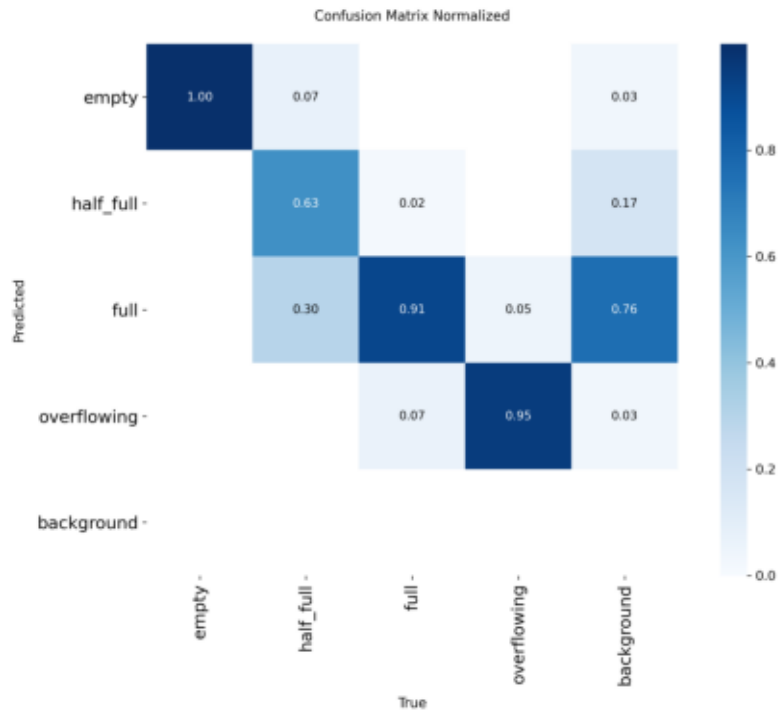


Validation Results Summary

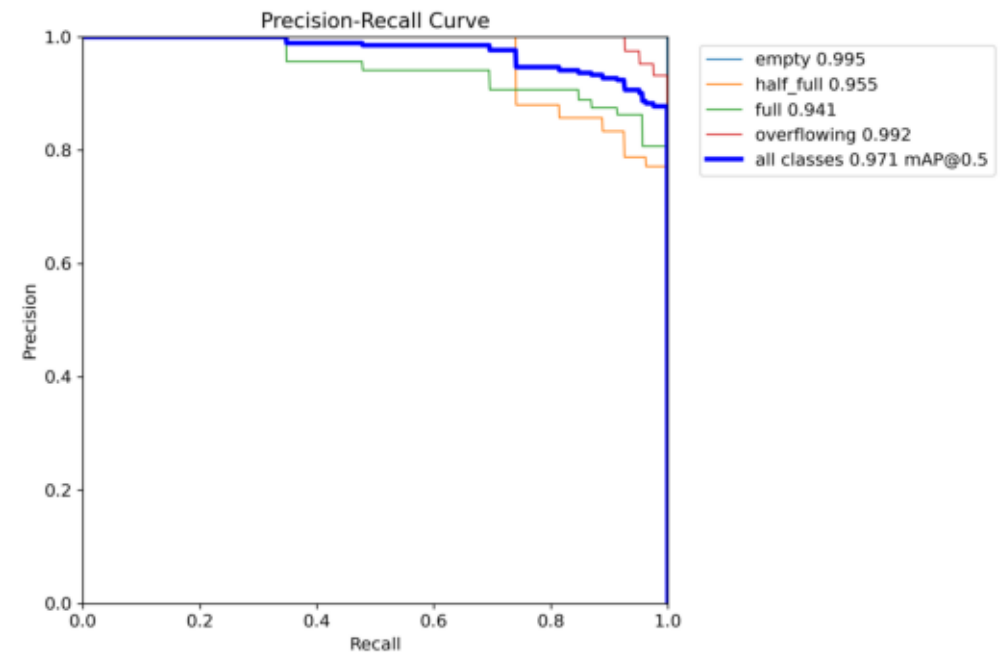
Class	Precision	Recall	mAP50	Images
Empty	0.629	1.000	0.995	6
Half Full	0.988	0.741	0.955	27
Full	0.659	1.000	0.941	46
Overflowing	0.952	0.975	0.992	41
OVERALL	0.807	0.929	0.971	120

Confusion Matrix & PR Curve

Normalised Confusion Matrix
(rows = predicted, cols = true)



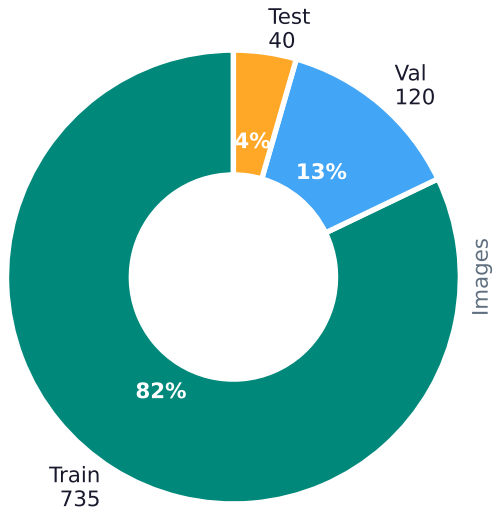
Precision-Recall Curve (all classes)



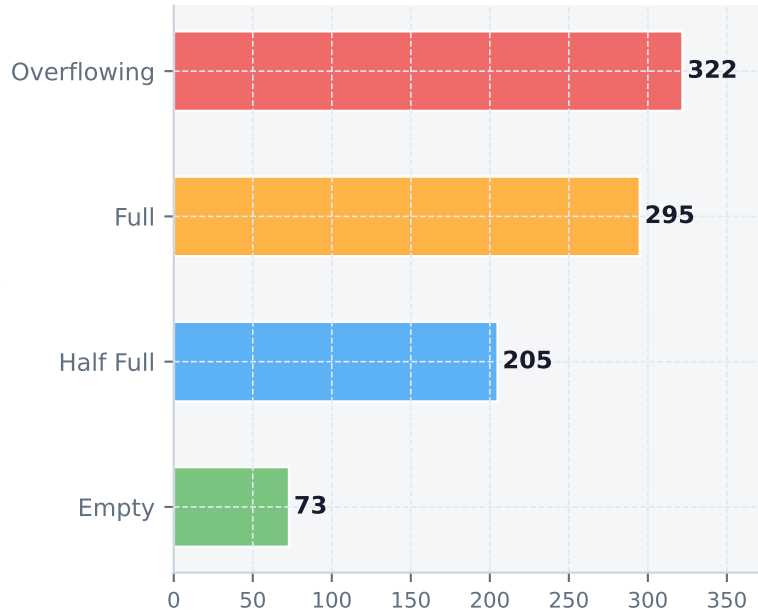
Empty → predicted empty: 1.00 · Overflowing → correct: 0.95 · No dangerous under-estimation

Dataset Profile & Sample Predictions

Train / Val / Test Split
(Device-based)



Dataset Class Distribution



Key Findings

- ✓ **mAP50 = 0.971**
Exceeds 0.80 target by 21pp
- ✓ **Empty recall = 1.00**
Zero missed empty bins
- ✓ **No dangerous errors**
Full/overflow never predicted empty
- ✓ **Unseen camera test set**
UKConnect device held out entirely
- ✓ **62 epochs, 15 min**
RTX 4050 GPU, early-stop at ep 42
- ✓ **895 real IoT images**
Proprietary industrial dataset

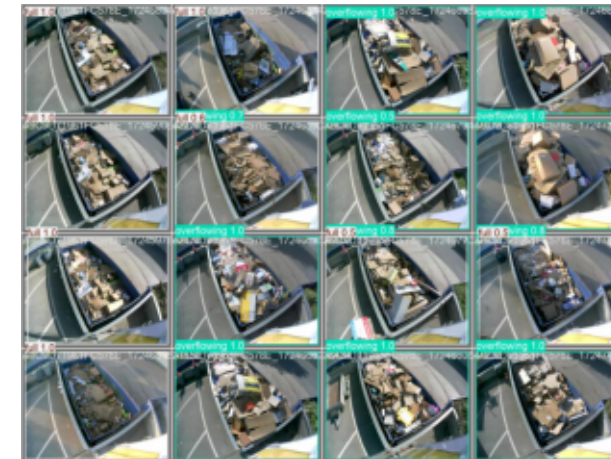
Validation Batch 1: Predictions



Validation Batch 2: Predictions



Validation Batch 3: Predictions



Go / No-Go Gate Results

Operational Significance

◆ Domain Adaptation

Existing model trained on small coloured plastic bins (mAP 0.85).
New model trained on large industrial steel bins, a visually distinct domain, achieving mAP 0.971 without any degradation.

◆ Safety-Critical Bias

Full and Overflowing bins are never misclassified as Empty.
The model over-collects rather than under-collects; the correct operational bias for a waste management system.

◆ No Manual Labelling

Fill percentage is embedded in the IoT sensor filename convention.
895 labelled images obtained without any manual annotation by parsing the proprietary naming schema.

◆ Production Integration

Deployed via FastAPI hot-swap endpoint: admin uploads new .pt file, backend runs test inference, promotes atomically.
Zero downtime. Existing API contracts unchanged.

Metric	Threshold	Achieved	Status
Overall mAP50	≥ 0.80	0.971	✓ PASS
Empty mAP50	≥ 0.75	0.995	✓ PASS
Half Full mAP50	≥ 0.75	0.955	✓ PASS
Full mAP50	≥ 0.75	0.941	✓ PASS
Overflowing mAP50	≥ 0.75	0.992	✓ PASS
Dangerous errors	= 0	None	✓ PASS